TRAINING DATASET

10,000+ such examples

Natural language question:

"Show all students with grade higher than 90"

Database schema (with comments and examples):

```
CREATE TABLE students (
    student_id INTEGER PRIMARY KEY,  -- Unique identifier
    name VARCHAR(100),               -- Student name
    grade INTEGER                    -- Grade (0-100)
);
```

Target SQL query:

```
SELECT name, grade FROM students WHERE grade > 90
```

MODEL ADAPTATION METHODS

FINE-TUNING

- Model weights update
- 10,000+ examples
- Epochs: 3-10
- GPU cluster
- Days/weeks of training

FEW-SHOT LEARNING

- No weights training
- 3-5 examples in prompt
- In-context learning
- Fast application
- Seconds/minutes

Base LLM Models: GPT-4, Claude 3.5, Gemini 2.0, Llama 3, Qwen

EVALUATION

CONTENT MATCHING BASED

- Component matching (CM)
- Exact matching (EM)

EXECUTION BASED

- Execution Acuracy (EA)
- Valid Efficiency Score(VES)

⚠ CRITICAL DEPENDENCY: DATASET QUALITY

Incorrect dataset → Poor training → Wrong SQL

Types of dataset errors:

- ❌ Structural: FK→non-existent tables, duplicate columns
- ❌ SQL syntax: parsing or execution erros
- ❌ Semantic: SQL syntactically OK, but logically ≠ question
- ❌ SQL anti-patterns: techically correct with performance issue